The examination is being carried out on the **following application documents**

Description, Pages

1-11 as originally filed

Claims, Numbers

1-15 as originally filed

Drawings, Sheets

1/6-6/6 as originally filed

Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure.

D1 US 5 224 234 A (ARSENAULT JEFFREY L [US] ET AL) 6 July 1993 (1993-07-06)

D2 US 6 546 585 B1 (BLAUSTEIN LAWRENCE A [US] ET AL) 15 April 2003 (2003-04-15)

D3 US 2019/246780 A1 (JUNGNICKEL UWE [DE] ET AL) 15 August 2019 (2019-08-15)

1 Novelty (Art. 52(1) and 54 EPC)

The present application does not meet the requirements of Article 52(1) EPC because the subject-matter of claim 1 is not new within the meaning of Article 54(1) and (2) EPC.

1.1 Document D1 discloses (references in brackets apply to this document):

A toothbrush (embodiment of figs. 22-24) including a handle (16) forming a gripping portion of the toothbrush and a replaceable brush head (81) carrying a plurality of bristles, wherein the replaceable head (81) is connectable to the handle (16) in a detachable manner by means of a snap locking mechanism (80, 82, 84, 86),

the snap locking mechanism including a finger (82) protruding from the handle (16) and a recess (80) formed in the brush head (81), wherein the finger (82) extends along a longitudinal direction (see fig. 24);

wherein the finger (82) includes a flexible portion (fig. 24, tip of finger 82 needs to elastically deform to pass over protrusion 86) and a non-flexible portion (fig. 24, rest of finger 82 after recess 84, does not deform in anyway) that are configured to be at least partly received in the recess (80) when the brush head (81) is mounted on the handle (16) (see figs. 22-24);

wherein the non-flexible portion (finger 82 without the tip) connects the flexible portion (tip of finger 82) to the handle (16) (see fig. 24);

wherein the flexible portion (tip of finger 82) is configured to elastically flex away from the bristles with respect to the non-flexible portion (finger 82 without the tip) in a direction transversal to the longitudinal direction during the insertion of the finger (82) into the recess (80) along the longitudinal direction (figs. 22 to 24 - when the finger 82 is mounted in recess 80, the tip flexes away from the bristles when it runs over the protrusion 82),

wherein the flexible portion (tip of finger 82) is configured to flex due to the interaction with an inclined surface (fig. 23 inclined surface of protrusion 86) formed in the recess (80) during insertion of the finger (82) into the recess (80) along the longitudinal direction,

wherein the flexible portion (tip of finger 82) includes a protrusion (tip of finger 82) configured to snap into an engagement position with the recess (80) when the protrusion (tip of finger 82) slides beyond the inclined surface (86) inside the recess (80).

The subject-matter of claim 1 is therefore not new over D1 (Article 54(1) and (2) EPC).

1.2 Document D2 also discloses all the features of claim 1:

A toothbrush (10) including a handle (12) forming a gripping portion of the toothbrush and a replaceable brush head (16) carrying a plurality of bristles (20), wherein the replaceable head (16) is connectable to the handle (12) in a detachable manner by means of a snap locking mechanism (42, 46, 48, 18, 19),

the snap locking mechanism including a finger (40) protruding from the handle (12) and a recess (18) formed in the brush head (16), wherein the finger (40) extends along a longitudinal direction (74) (see fig. 2-4);

wherein the finger (40) includes a flexible portion (42) and a non-flexible portion (24, base of finger 40) that are configured to be at least partly received in the recess (18) when the brush head (16) is mounted on the handle (12) (see figs. 3B and 4);

wherein the non-flexible portion (24) connects the flexible portion (42) to the handle (12) (see fig. 3);

wherein the flexible portion (42) is configured to elastically flex away from the bristles with respect to the non-flexible portion (24) in a direction transversal to the longitudinal direction during the insertion of the finger (40) into the recess (18) along the longitudinal direction (figs. 3 and 4, column 4 lines 7 to 25),

wherein the flexible portion (42) is configured to flex due to the interaction with an inclined surface (inclined surface inside aperture 18; opposite surface 62; see dotted line in fig. 3) formed in the recess (18) during insertion of the finger (40) into the recess (18) along the longitudinal direction,

wherein the flexible portion (42) includes a protrusion (46) configured to snap into an engagement position with the recess (18) when the protrusion (46) slides beyond the inclined surface (inclined surface inside aperture 18) inside the recess (18) (figs. 3 and 4 show that the protrusion 46 slides on the inclined surface of aperture 18 until it snaps in aperture 19).

Therefore, the subject-matter of claim 1 is also not new over D2 (Article 54(1) and (2) EPC).

1.3 Claim 1 is so broad that it allows for the very different configuration of the snap locking mechanism of document D3 to be read into this claim:

A toothbrush (10) including a handle (12) forming a gripping portion of the toothbrush and a replaceable brush head (16) carrying a plurality of bristles, wherein the replaceable head (16) is connectable to the handle (12) in a detachable manner by means of a snap locking mechanism (42, 70, figs. 8 and 11),

the snap locking mechanism including a finger (14) protruding from the handle (12) and a recess (66) formed in the brush head (16), wherein the finger (14) extends along a longitudinal direction (see fig. 3);

wherein the finger (14) includes a flexible portion (42) and a non-flexible portion (18, base of finger 14) that are configured to be at least partly received in the recess (18) when the brush head (16) is mounted on the handle (12) (see fig. 1);

wherein the non-flexible portion (18) connects the flexible portion (42) to the handle (12) (see figs. 1, 8);

wherein the flexible portion (42) is configured to elastically flex away from the bristles with respect to the non-flexible portion (18) in a direction transversal to the longitudinal direction during the insertion of the finger (14) into the recess (66) along the longitudinal direction (figs. 8 and 11, paragraph [26]),

wherein the flexible portion (42) is configured to flex due to the interaction with an inclined surface (fig. 11, inclined surface inside aperture 66) formed in the recess (66) during insertion of the finger (14) into the recess (66) along the longitudinal direction (paragraph [26]),

wherein the flexible portion (42) includes a protrusion (ball 44) configured to snap into an engagement position with the recess (18) when the protrusion (44) slides beyond the inclined surface (fig. 11, ball 44 slides inclined surface inside aperture 66 and snaps in recess 70) inside the recess (66) (figs. 8 and 11).

Therefore, the subject-matter of claim 1 is also not new over D3 (Article 54(1) and (2) EPC).

1.4 **Dependent claims 2 to 14** do not appear to contain any additional features which, in combination with the features of any claim to which they refer, meet the requirements of the EPC with respect to novelty and/or inventive step:

1.4.1 Claims 2 and 7 define the recess being open on the back side of the head and the finger having a T-shaped cross-section at the level of the non-flexible portion. D1 discloses the open back (see fig. 23) and both D1 and D2 disclose a T-shaped cross-section for the non-flexible portion (see fig. 24 for D1 and fig. 3A for D2.)

1.4.2 Claims 3 to 6 and 9, 10 define dimensions of the snap mechanism. The subject-matter of these claims consists in the selection of values from known ranges. Such a selection can only be regarded as inventive if these dimensions presents unexpected effects or properties in relation to the rest of the range. However, no such effects or properties are indicated in the application.

1.4.3 Claims 8 and 11, 12, 13 define 2 lateral ribs on the non-flexible portion to engage into corresponding grooves formed in the recess. Both D1 (figs. 23, 24) and D2 (fig. 3) have lateral extension on the finger to engage with the recess. D1 also has corresponding grooves in the recess 80 (see fig. 23).

1.4.4 Claim 14 defines a concavity between the non-flexible portion and the protrusion on an upper side of the finger. D1 also discloses concavity 82 (see fig. 24). Furthermore, this is a common feature for a snap connection.

1.4.5 Claim 15 defines the longitudinal direction (X) of the finger is substantially parallel to a longitudinal direction of a gripping portion. Both D1 (fig. 3 and 4) and D2 (fig. 4) disclose the same.

1.5 Additional remarks on novelty

A combination of claim 1 and these additional features: the open back, the non-flexible portion and the flexible portion being integral to each other, the non-flexible portion being fixed via ribs in the open back recess and the flexible portion being free to flex out of the open back because it does not have ribs, appears to be novel over the cited prior art. See description page 3, lines 27-32, page 7, lines 22, 23, page 9, lines 9-16.

It is noted that this suggestion is made only for assisting the applicant in deciding how to proceed. It in no way precludes consideration of alternative solutions submitted by the applicant. The responsibility for determining the text of the application (Article 113(2) EPC) and, in particular, for defining the subject-matter for which protection is sought remains with the applicant.

2 Procedure

2.1 The applicant is invited to file new claims which take account of the above comments.

2.2 In order to comply with the requirements of Rule 137(4) EPC, the applicant should clearly identify the amendments made, irrespective of whether they concern amendments by addition, replacement or deletion, and indicate the passages of the application as filed on which these amendments are based (see Guidelines H III, 2.1).

If the applicant considers it appropriate, these indications could be submitted in machine written form (hand-written amendments are no longer accepted as pointed out om OJ EPO 2013, 603) on a copy of the relevant parts of the application as filed or in documents with tracked changes.

2.3 In order to assess the question of inventive step, the applicant is asked to indicate in the response which technical problem is solved by the characterising features of the new independent claim compared to the closest prior art (Rule 42(1)(c) EPC).

2.4 In case the applicant accepts the suggestion of the examiner for an acceptable form of amendments to the claims overcoming the objections raised, the applicant should also submit an adapted description together with the amended claims.

Any inconsistency between the description and the claims which may throw doubt on the extent of protection must be avoided to fulfil the requirements of Article 84 EPC. Article 69 EPC provides no grounds for dispensing with amendment of the description.

Requirements for the adaptation of the description:

– The technical field of the invention (Rule 42(1)(a), GL F-II.4.2) must correspond to the amended set of claims.

– The summary of the invention (Rule 42(1)(c) and GL F-II.4.5) must correspond to the amended set of claims. If appropriate, statements such as “The invention is set out in the appended set of claims“ or “The invention is as defined in claim X“ can be applied instead of repeating the claims verbatim.

– All technical features of the independent claims shall not be presented as optional in the description. Terms such as “for example“, “may“, “can”, “exemplary“, “optionally“, “preferably“ and the like should be removed when preceding a feature of an independent claim.

– The expression "incorporated herein by reference" (or similar) should be deleted in each and every instance (see Guidelines F-III,8 and Art. 84 EPC).

– The closest prior art documents D1 and D2, and the differences between their teachings and the subject-matter of the independent claims currently on file, should be briefly summarised in the description (see Guidelines F-II, 4.3 and Rule 42(1)(b) EPC).